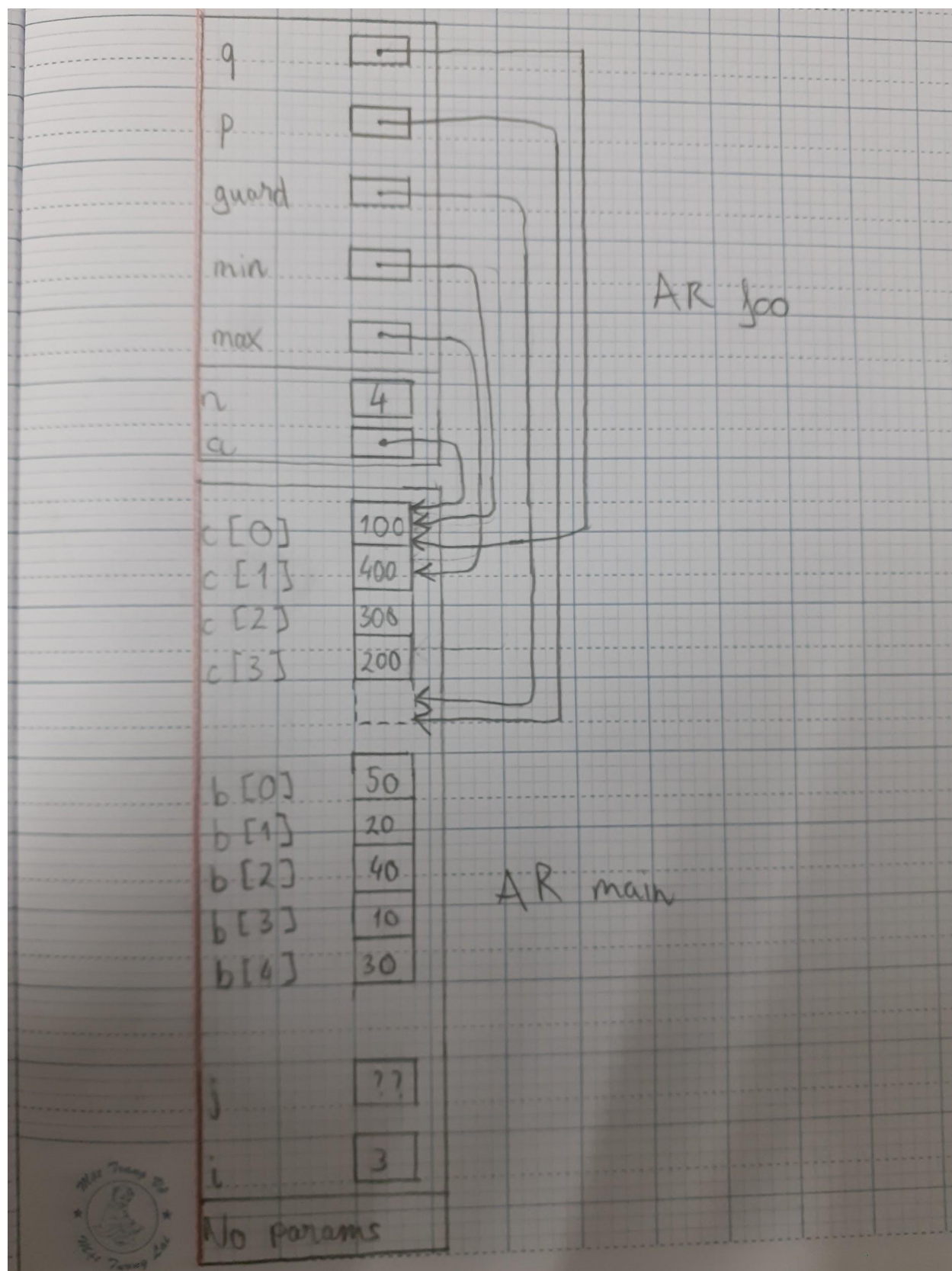


Duc Nguyen - UCID: 30102065

Lab section B01

ENCM 335 - LAB 6

Exercise B:



Exercise C:

Code:

```
1 // charIO4C.c
2 // ENCM 335 Fall 2021 Lab 6 Exercise C
3
4 #include <stdio.h>
5 #include <ctype.h>
6
7 #define QUIT_LETTER 'q'
8
9 int main(void)
10 {
11     int c, line_length, num_letters, num_digits;
12
13     // Each pass through the outer loop reads a line of input.
14     while (1) {
15         printf("\nEnter a line of text. (To quit, start the line with %c.)\n",
16             QUIT_LETTER);
17         c = fgetc(stdin);
18         if (c == EOF || c == QUIT_LETTER)
19             break; // Leave outer loop.
20
21         line_length = 0;
22         num_letters = 0;
23         num_digits = 0;
24         printf("Changing uppercase to lowercase ...\n");
25
26         // Inner loop reads chars until '\n' or EOF is found.
27         while (c != '\n' && c != EOF) {
28             line_length++;
29             if (isalpha(c))
30                 num_letters++;
31
32             else if (isdigit(c))
33                 num_digits++;
34             fputc(tolower(c), stdout);
35             c = fgetc(stdin);
36         }
37         fputc('\n', stdout);
38         printf("Length of line, not counting '\\n', was %d.\n", line_length);
39         printf("%d of the characters in the line were letters.\n", num_letters);
40         printf("%d of the characters in the line were digits.\n", num_digits);
41     }
42     printf("Bye!\n");
43     return 0;
44 }
```

Output:

```
$ ./a.exe
Enter a line of text. (To quit, start the line with q.)
2019 version of ENCM 335
Changing uppercase to lowercase ...
2019 version of encm 335
Length of line, not counting '\n', was 24.
13 of the characters in the line were letters.
7 of the characters in the line were digits.

Enter a line of text. (To quit, start the line with q.)
q
Bye!
```

Exercise D:

Part 1:

```
$ ./a.exe
Please enter a line of text. To quit, start it with q.
abcde

Input line was "abcde"
Please enter a line of text. To quit, start it with q.
abcdefgfh

Input line was "abcdefgfh"
Please enter a line of text. To quit, start it with q.
abcdefgghi

Input line was "abcdefgghi"
Please enter a line of text. To quit, start it with q.

Input line was ""
Please enter a line of text. To quit, start it with q.
aaaaaaaaaaaaaaaaaaaaa

Input line was "aaaaaaaaaaaa"
Please enter a line of text. To quit, start it with q.

Input line was "aaaaaaaaaaaa"
Please enter a line of text. To quit, start it with q.

Input line was "a"
Please enter a line of text. To quit, start it with q.
q

Reason for quitting: found q at beginning of line.
```

Part 2:

Code:

```
1 // line-getter6D.c
2 // ENCM 335 Fall 2021 Lab 6 Exercise D
3
4 #include <stdio.h>
5 #include <string.h>
6
7 #define QUIT_LETTER 'q'
8
9 // Again, in a practical program, this is a ridiculously small size
10 // for an array that is supposed to hold a line of text. But it's
11 // convenient for testing purposes.
12 #define LINE_ARRAY_SIZE 10
13
14 int eat_til_newline(FILE *stream);
15 // REQUIRES: stream is open for input.
16 // PROMISES: Characters are read from stream and discarded until either a
17 // '\n' has been read or an input error has occurred.
18 // Return value is 0 if '\n' is read, and EOF for an error.
19
20 int get_a_line(char *s, int size, FILE *stream);
21 // Does what fgets does, using repeated calls to fgets, but
22 // provides a more useful return value than fgets does.
23 //
24 // REQUIRES
25 //   size > 1.
26 //   s points to the start of an array of at least size bytes.
27 //   stream is open for input.
28 // PROMISES
29 //   Return value is EOF if input error occurred.
30 //   Otherwise, return value gives the index of the '\0' that
```

```

31 // terminates the string in the array.
32
33 void reverse(char *s);
34 // REQUIRES
35 //   s points to the start of an array of at least size bytes.
36 //   stream is open for input.
37 // PROMISES
38 //   Reverse all values of the input string
39
40 int main(void)
41 {
42     char line[LINE_ARRAY_SIZE];
43     int end_line;
44
45     while (1){
46         printf("Please enter a line of text. To quit, start it with %c.\n",
47             QUIT_LETTER);
48         end_line = get_a_line(line, LINE_ARRAY_SIZE, stdin);
49         if (line[0] == QUIT_LETTER)
50             break;
51         if (end_line < LINE_ARRAY_SIZE - 1){
52             printf("The line, newline removed, was \"%s\".", line);
53             reverse(line);
54             printf(" In reverse, that is \"%s\".\n", line);
55         }
56         else if (end_line >= LINE_ARRAY_SIZE - 1){
57             fputs("Input line ignored because it was too long!\n", stdout);
58             eat_til_newline(stdin);
59         }
60     }

```

```

61
62     fputs("\nReason for quitting: ", stdout);
63     if (end_line == EOF)
64         fputs("unexpected input error.\n", stdout);
65     else
66         printf("found %c at beginning of line.\n", QUIT_LETTER);
67
68     return 0;
69 }
70
71 int eat_til_newline(FILE * stream)
72 {
73     int c;
74     do {
75         c = fgetc(stream);
76     } while (c != EOF && c != '\n');
77
78     // Return EOF if c == EOF, otherwise return 0.
79     return (c == EOF) ? EOF : 0;
80 }
81
82 void reverse(char *s)
83 {
84     // You need to add some code here!
85     int i, len, holder;
86     len = strlen(s);
87     for (i = 0; i < len / 2; i++){
88         holder = s[i];
89         s[i] = s[len - i - 1];
90         s[len - i - 1] = holder;

```

```

91     }
92 }
93 int get_a_line(char *s, int size, FILE *stream){
94     int i;
95     for (i = 0; i < size - 1; i++){
96         s[i] = fgetc(stream);
97         if (s[i] == EOF)
98             return EOF;
99         if (s[i] == '\n')
100             break;
101     }
102     s[i] = '\0';
103
104     return i;
105 }
106 }
107

```

Output:

```
$ ./a.exe
Please enter a line of text. To quit, start it with q.
abcde
The line, newline removed, was "abcde". In reverse, that is "edcba".
Please enter a line of text. To quit, start it with q.
abcdefgh
The line, newline removed, was "abcdefgh". In reverse, that is "hgfedcba".
Please enter a line of text. To quit, start it with q.
abcdefghi
Input line ignored because it was too long!
Please enter a line of text. To quit, start it with q.
aaaaaaaaaaaa
Input line ignored because it was too long!
Please enter a line of text. To quit, start it with q.
q
Reason for quitting: found q at beginning of line.
```